NAU NORTHERN ARIZONA UNIVERSITY
Steve Sanghi College of Engineering

Memorandum

MA2 - Conditionals, Timing, and Functions

to: course staff

from: project team

context: undergraduate engineering project

Re: MA2

1.1) Why is it problematic to use a delay() statement in your control code?

It is problematic to use a delay() statement in control code because it can block other functions of the code from running. This can block events that need to run on strict timing, or real-time sensor checks.

1.2)What is the difference between period and frequency?

Period is the time it takes for one event cycle to finish, usually measured in seconds. Frequency is how many cycles per second, measured in Hertz (Hz).

1.3) See figure one.

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Figure 1 – The output from the serial monitor for program MA2_1. The 10 Hz statement occurs 10 times for every one 1 Hz statement

2.1) If a user enters "123", only "1" is printed to the monitor. Why?

Only "1" is printed to the monitor because the Serial.read() function only returns the first byte of serial data. After data is read, the ClearSerialBuffer() function "throws away" the rest of the serial data, so no other characters can be output. A small delay() was required before "throwing away" the data, as without it, inputs with more than one character had every character print, because the loop was running too quickly for the buffer to be properly dumped.

2.2) What is necessary to include in a function declaration?

In a function declaration, it is necessary to include the return type of the function (void, int, bool), the function name, and any parameters, if applicable, and their variable types.

3.1) What are the challenges of replacing the entire if statement with a function?

The challenges of replacing the entire if statement with a function are that updating the variables like prevTime1 and prevTime2 still must happen outside of the function, so there is not much code saved or readability improvement compared to the original MA2_1. Additionally, the period of the 10 Hz and 1 Hz loops must get recalculated with every call of the function, while the original program stored the periods as constants. This introduces performance degradation, but it is a minor effect.